

.....(1).....
MECHANICS

$$\vec{v} = \frac{ds}{dt}$$

$$\vec{a} = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

Where $\bar{v}$ is the average velocity:

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

Assuming acceleration is constant:

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\vec{x}(t) = \vec{x}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2$$

$$\vec{x}(t) = \vec{x}_0 + \frac{\vec{v}_0 + v}{2}t$$

$$||v|| = \sqrt{v_0^2 + 2||a|| ||x - x_0||}$$

.....(2).....
FORCES

$$\vec{F}_g = m\vec{g}$$

$$||\vec{F}_{sp}|| = k \Delta \ell$$

$$||f_s|| \leq ||\mu_s \vec{N}||$$

$$\vec{f}_k = \mu_k \vec{N}$$

Newton's laws:

1. A body remains at rest, or in motion at a constant speed in a straight line, unless it is acted upon by a force.
2. $\sum \vec{F} = m\vec{a}$
3. If two bodies exert forces $\vec{F}_1, \vec{F}_2$ on each other, then $\vec{F}_1 = -\vec{F}_2$.

.....(3).....
ENERGY

$$W = \int \vec{F}(s) ds$$

$$(\exists c, \theta \forall t: |F(t)| = c, \angle F(t) = \theta)$$

$$\Rightarrow W = F_x \cdot \Delta x = F \cos \theta \Delta s$$

$$E_k = \frac{1}{2}m||v||^2$$

$$U_g = mgh$$

$$U_{sp} = \frac{1}{2}k(\Delta \ell)^2$$

$$E_{k1} + U_{g1} = E_{k2} + U_{g2}$$

$$\forall i, j: (\sum E)_i = (\sum E)_j$$

$$W_F = \Delta E = E_{\text{final}} - E_{\text{beginning}}$$

.....(4).....
ROTATIONAL MOVEMENT

$$f := \frac{1}{T} \quad [\text{Hz}]$$

$$L = r\psi_{\text{rad}}$$

$$\omega := 2\pi f = \frac{2\pi}{T}$$

$$v = \frac{2\pi r}{T}$$

$$\bar{\omega} = \frac{\Delta \theta}{\Delta t}$$

$$v = \omega r$$

$$||\vec{a}_R|| = \frac{||v||^2}{r} = \omega^2 r$$

$$P = 2\pi r$$

Critical Velocity at max.:

$$N = 0 \iff v = \sqrt{gr}$$

$$\vec{a}_T = -g \sin \alpha$$

$$\vec{a} := \vec{a}_T + \vec{a}_r$$

$$||a|| = \sqrt{a_T^2 + a_R^2}$$

$$\tan \theta = \frac{|a_T|}{|a_R|}$$

.....(5).....
GRAVITY

$$\left(\frac{\bar{r}_1}{\bar{r}_2}\right)^3 = \left(\frac{T_1}{T_2}\right)^2$$

$$F_g = G \frac{Mm}{r^2}$$

$$U_G = -\frac{GMm}{r}$$

$$(W_g)_{A \rightarrow B} = GMm \left(\frac{1}{r_A} - \frac{1}{r_B} \right)$$

$$\rho = \frac{m}{v}$$

In a circular motion:

$$E_k = \frac{GMm}{2r} = -\frac{U_G}{2}$$

$$\sum E_{\text{mechanic}} = E_k + U_G$$

$$= -\frac{GMm}{2r} = -E_k$$

$$T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{GM}}$$

Kepler's laws of planetary motion:

1. The orbit of a planet is an ellipse
2. A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time.
3. For a given gravitational system:

$$\exists c \forall i: \frac{T_i^2}{r_i^3} = c = \frac{4\pi^2}{GM}$$

.....(6).....
MOMENTUM

$$\vec{p} = m\vec{v} \quad [N \text{ sec}] \sim \left[\frac{\text{kg m}}{\text{sec}} \right]$$

$$\vec{J} = \int F dt \stackrel{!}{=} \vec{F} \cdot \Delta \vec{t} = |\vec{F}| |\Delta \vec{t}| \cos \theta$$

$$\vec{J}_{\Sigma F} = \sum_{i=1}^n \vec{J}_{F_i} = \Delta \vec{p}$$

$$\forall t_1, t_2 \in \mathbb{R}: \sum_{i=1}^n \vec{p}_i(t_1) = \sum_{i=1}^n \vec{p}_i(t_2)$$

In an inelastic collision:

$$Q = \Delta E_k$$

In an elastic collision, where v_i before collision and u_i after it:

$$\vec{v}_1 - \vec{v}_2 = -(\vec{u}_1 - \vec{u}_2)$$

Elastic Collision iff no loss of kinetic energy, Inelastic iff not ecstatic.

.....(7).....
CONSTANTS

$M [kg]$	$R [m]$	Obj
$5.974 \cdot 10^{24}$	$6.38 \cdot 10^6$	Earth
$1.99 \cdot 10^{30}$	$6.96 \cdot 10^8$	Sun
$7.35 \cdot 10^{22}$	$1.74 \cdot 10^6$	Moon

$$G = 6.67 \cdot 10^{-11} \frac{\text{Nm}^2}{\text{kg}}$$

Obj	Mass [kg]	Charge [C]
Electron	$9.11 \cdot 10^{-31}$	$-1.6 \cdot 10^{-19}$
Proton	$1.67 \cdot 10^{-27}$	$+1.6 \cdot 10^{-19}$
Neutron	$1.67 \cdot 10^{-27}$	0

$$k \approx 9 \cdot 10^9 \left[\frac{Nm^2}{C^2} \right] = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 := \frac{1}{4\pi k} \approx 8.85 \cdot 10^{-12}$$

.....(8).....
ELECTRICITY

Coulomb's law:

$$\left\| \vec{F}_E \right\| = k \frac{|q_1| |q_2|}{r^2}$$

Definition of the electrical field of a given particle with charge q :

$$\left\| \vec{E} \right\| = \frac{\vec{F}_E}{q_B} = k \frac{q}{r^2} \left[\frac{N}{C} \right]$$

Definition of charge density:

$$\sigma = \frac{Q}{A}$$

Given an infinity large plane:

$$E = 2\pi k \sigma = \frac{\sigma}{2\epsilon_0}$$

$$\begin{aligned} \Delta W_{A \rightarrow B} &= k q_1 q_2 \left(\frac{1}{r_B} - \frac{1}{r_A} \right) \\ &= (U_E)_B - (U_E)_A \end{aligned}$$

Hence the potential electric energy:

$$U_E = k \sum_{A,B} \frac{q_A q_B}{\text{dist}(A,B)} \left(\text{singular } \frac{kqQ}{r} \right)$$

Definition of electric potential in a given point:

$$V_A = \sum_B k \frac{q_B}{\text{dist}(A,B)} \left[V = \frac{J}{C} \right]$$

$$W_{A \rightarrow B} = q V_{AB} = q \underbrace{(V_B - V_A)}_{\text{Voltage}}$$

$$E = -\frac{dv}{dx} \quad V = -\int E dx \stackrel{!}{=} \frac{E}{d}$$

Potential inside a conductive ball:

$$V = \frac{kQ}{R}$$

.....(9).....
CAPACITORS

Where V is the voltage between both ends of the capacitor, the capacitance is defined as:

$$C := \frac{q}{V} \left[F = \frac{Q}{V} = \frac{J}{V^2} \right] \implies q = CV$$

$$V = E \cdot d$$

$$U_{\text{capacitor}} = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

$$U_{\text{battery}} = QV = CV^2 = \frac{Q^2}{C}$$

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$E = \frac{V}{d} = \frac{\sigma}{\epsilon_0 \epsilon_r} = \frac{Q}{A \epsilon_0 \epsilon_r}$$

But for circular capacitor:

$$C = \frac{R_{\text{inner}} R_{\text{outer}}}{k(R_{\text{outer}} - R_{\text{inner}})}$$

(needs proving) when a capacitor is unconnected to a voltage source, inserting a dielectric will require:

$$W_{\text{external}} = \frac{Q^2}{2C} \left(\frac{1}{\epsilon_r} - 1 \right) < 0$$

When connected to a battery:

$$W_{\text{external}} = -\frac{1}{2} V^2 C (\epsilon_r - 1) < 0$$

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Shahar Perets, 2025

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